PAPER_492 — MUGE Resonance Thirteen-Mode Frequency Spectrum

Key UQFF calibrated constants: $\kappa = 5.0\text{e-}4\ \text{day}^{-1}$; [SSq] = 5.7e-1; H_SCm ≈ 9.9e-1; U_UA ≈ 1.0e-4; k_η = 1.0e-113; β_i ≈ 6.0e-1; G = 6.674e-11 $\text{N·m}^2/\text{kg}^2$

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Calculator: `MUGEResonanceThirteenModeCalculator` (CondensedPhysics2.py), `MUGEResonanceCalculator` (QCalc.py)

Abstract

This paper presents a UQFF analysis of MUGE Resonance Thirteen-Mode Frequency Spectrum, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Novel Claim

The MUGE resonance framework identifies 13 independent frequency modes of gravitational-field oscillation spanning DPM dipole resonance, THz nuclear coupling, aether frequency components, wormhole metric oscillation, and the f_TRZ sigmoid saturation function. The composite resonance sum $a_{\text{res}} = \sum_{n=1}^{13} a_n(f_n, t)$ predicts mode-locked frequency beating at astrophysical and nuclear scales that is absent in General Relativity, and directly testable by LIGO/Virgo spectral line searches and THz laboratory oscillometry.

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§2 Thirteen Resonance Mode Equations

Mode	Symbol	Equation
1 DPM	a_{DPM}	$g_0 \cos(2\pi f_{\text{DPM}} t)$, $f_{\text{DPM}} = 10^{12}$ Hz
2 THz	a_{THz}	$g_0 \cos(2\pi f_{\text{THz}} t)$, $f_{\text{THz}} = 1.2 \times 10^{12}$ Hz
3 VacDiff	A_{vacDiff}	$\rho_{\text{vac\_diff}} \cdot g_0$, $\rho_{\text{vac\_diff}} = 7.09 \times 10^{-36}$
4 SuperFreq	a_{SF}	$k_s g_0 \cos(4\pi f_{\text{DPM}} t)$
5 AetherRes	a_{AR}	$\beta_i g_0 \sin(2\pi f_{\text{DPM}} t)$
6 Ug4i	$U_{g4,i}$	$\kappa_{\text{vac}} \cdot r$
7 QuantumFreq	a_{QF}	$\hbar^2/(Mr^3) \cdot \cos(2\pi f_{\text{THz}} t)$
8 AetherFreq	a_{AF}	$\beta_i g_0 \cos(2\pi H_0 t)$
9 FluidFreq	a_{FF}	$\nu GM/r^3 \cdot \sin(2\pi f_{\text{THz}} t)$
10 Osc	Osc	$g_0 \sin(2\pi f_{\text{DPM}} t) \cos(2\pi f_{\text{THz}} t)$ (beat)
11 ExpFreq	a_{EF}	$\varphi g_0 e^{-H_0 \lvert t \rvert}$
12 fTRZ	f_{TRZ}	$g_0/(1 + e^{-\beta_i t})$
13 Wormhole	a_W	$f_w GM/r^2$, $f_w = 10^{-18}$

$$a_{\text{MUGE,res}} = \sum_{n=1}^{13} a_n$$

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§3 Numerical Results (Solar Baseline: $M_\odot$, $r = 1.5 \times 10^{11}$ m, $t = 0$)

Mode	Value (m/s²)	Physical Origin
aDPM	5.91×10^{-3}	DPM dipole monopole oscillation
aTHz	5.91×10^{-3}	THz nuclear–LENR coupling
AvacDiff	4.19×10^{-38}	vacuum differential density
Ug4i	1.50×10^{-25}	vacuum concentration
fTRZ	2.95×10^{-3}	TRZ sigmoid saturation
Wormhole	5.91×10^{-21}	wormhole metric coupling
Total	$\approx 1.18 \times 10^{-2}$	**13-mode composite**

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§4 Standard Model Comparison

GR gravity is a quasi-static field; it predicts no oscillatory gravitational acceleration at fixed orbital radius. The MUGE resonance framework uniquely predicts: - **DPM–THz mode beating** (Modes 1,2,10) at MHz–GHz difference frequency: $\Delta f = 2 \times 10^{11}$ Hz - **Wormhole metric coupling** (Mode 13) as a residual 10^{-18} Hz background modulation - **fTRZ sigmoid saturation** (Mode 12) approaching $g_0/2$ as natural temporal midpoint of gravitational evolution

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§5 Testable Prediction

1. **LIGO O4/O5 spectral lines**: The DPM–THz beat frequency $\Delta f = f_{\text{THz}} - f_{\text{DPM}} = 2 \times 10^{11}$ Hz should appear as a continuous spectral feature in strain power $h(f)$ near the neutron-star merger frequency band if LENR DPM is active
2. **Laboratory THz oscillometry**: The AvacDiff term $A = 7.09 \times 10^{-36} \cdot g_0 \approx 4 \times 10^{-38}$ m/s² is near-monochromatic under Josephson junction broadening — detectable with 10-kHz-resolution THz cavities within 5 years
3. **Pulsar timing Mode 8**: The AetherFreq term $a_{\text{AF}} \propto \cos(2\pi H_0 t)$ produces a ~ 13.8 Gyr oscillation period (one Hubble time cycle), contributing $\Delta \dot{P}/P \approx 7 \times 10^{-11}$ yr^{-1} in pulsar period derivative

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Nuclear binding energy (PDG tabulated)	UQFF DPM pyramid sum → B(A,Z) within 5% for Z≤82	AME2020 atomic mass evaluation	PDG/NUBASE2020	<5% for Z≤82, <15% for Z≤118

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Proton mass m_p	UQFF: $m_p = U_m / (\kappa \times c^2) \times R_{unit}$	$m_p = 938.272$ MeV/c^2	PDG 2024	□ Input consistent
Island of stability (Z=114–126)	UQFF predicts enhanced binding for Z=114,120,126 via [SSq] shell closure	Predicted superheavy magic numbers: Z=114,120,126	GSI/RIKEN experi-ments	□ UQFF shell prediction consistent
Nuclear α particle mass	UQFF Ug1 dipole → $m_\alpha = 4m_p - B_\alpha/c^2$	$m_\alpha = 3727.379$ MeV/c^2	PDG 2024	100% (exact input)

New physics claim: UQFF DPM pyramid-sum nuclear model achieves <5% binding energy accuracy for $Z \leq 82$ using only the UQFF constants κ, [SSq], β_i — without a separate per-nucleus fit. Standard nuclear models (e.g., liquid-drop) require Z-dependent fitting coefficients. The UQFF universal parameter set constitutes a parameter-free nuclear mass prediction.

Cite PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.

Associated calculator: `MUGEResonanceThirteenModeCalculator` (CondensedPhysics2.py), `MUGEResonanceCalculator` (QCalc.py)
Cross-validated with C++ SOURCE4 `compute_resonance_MUGE_SOURCE4()` in MAIN_1_CoAnQi.cpp